Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech Mechanical Engineering (Thermal) Examination

November 2018

ME509 Design of Solar & Wind System

Date: 27.11.2018, Tuesday **Time: 01.30 p.m. To 04.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the question below.

(a) List out the methods for converting sun energy into electric power and discuss the difficulties presently faced by the same. **[07]**

(b) The latitude of Srinagar is 34^0C.Find day length in hrs When sun light is available on 1^{st} July. **[03]**

Q – 2.a A hot water storage tank system is used is to be designed to store 500 KWh energy. The maximum and minimum water temperatures are 85^0 C and 25^0 C. Find out the volume of water tank required for the purpose. Take properties of water at mean temperature of 55^0 C, C_{pw}= 4.18 KJ/kg ^{0}C and ρ= 988 kg/m^3 **[05]**

Q – 2.b Answer any two questions. **[10]**

(i) Derive an expression for useful heat gain rate of parabolic collector with cylindrical absorber of diameter d and length L. Assume that the heat transfer fluid change its phase as it passes through absorber and consequently remains at constant temperature T_{fs}.

(ii) In ideal absorption refrigeration system the evaporator temperature is 10^0C and condenser temperature is 30^0C. The system is run by using solar collector which can supply heat at 10^0C. Find the ideal COP of the system. If the load on the evaporator is 10 Tons of refrigeration, and actual COP is 50% of ideal, then find out the net heat supplied by the collector and if the solar radiation on the collector is 800 W/m^2 and net heat gain in the generator is 75% of the solar radiation, the find out the area of the solar collector required.

(iii) Discuss active and passive solar heating methods with neat sketch.

Q - 3 Answer any two questions. **[10]**

(i) Draw a neat diagram of absorption refrigeration system assisted by solar energy and explain its working.

(ii) Write a short note on LHTES.

(iii) A cylindrical parabolic collector is located keeping its focal axis west horizontal and the collector is rotated about the horizontal E-W axis adjusted continuously so that the solar beam makes minimum angle of incidence with aperture at all time. Find out

(a) Slope of aperture plane (b) Incident angle on 15th April where LAT = 12.30 hrs. Take following data at the place where collector is located latitude angle = 19.2^0 and hour angle = -7.5^0

SECTION – II

Q - 4 Answer the question below.

(i) What do u understood by concentration? Define and discuss the different terms which are important for the design of concentrating collector. **[07]**

Q – 5.a What do u understood by solar thermal energy storage? Explain why and when it is necessary. **[02]**

Q – 5.b Discuss Solar chimney power plant with sketch. **[07]**

OR

Q – 5.b An irrigation pump is to be designed to lift 8 m^3/min of water through a total head of 6.5 m. It is run by a turbine using working fluid F- 114. The turbine pressure is 10 bar and condenser pressure is 1.5 bar. The vapour leaving the turbine has 5^0C superheat and liquid coming out the condenser is saturated. Neglecting the losses and assuming the expansion is isentropic find out **[07]**

(a) the mass of F-114 circulated through the cycle

(b) the power cycle efficiency ρ_w= 1200 kg/m^3 and C_{pw}= 5.2 KJ/Kg^{0}C and temperature difference = 8^0C

enthalpy at inlet = 385 KJ/Kg, condenser inlet = 355 KJ/Kg and pump inlet and outlet is 210KJ/kg

Q – 5.c Discuss in detail the operation and control of wind turbine. **[05]**

Q – 6.a Explain the momentum theory in wind power generation. Further prove that the maximum output from turbine can be achieved by $V_e = \frac{Vi}{3}$ **[07]**

Q – 6.b At particular site where atmospheric pressure is 1.01325 bar and temp is 20^0C, the wind is an available at 8 m/s. Find out the following **[07]**

(a) power density available in the wind

(b) Max power density possible

(c) Obtainable power density assuming the overall efficiency is 35%

(d) Power capacity of the wind mill if its diameter is 30 m

OR

Q – 6.a Draw the neat diagrams of savonious and darrieus wind mills and explain their working. Discuss their merits and applications. **[07]**

Q – 6.b The wind velocity at a wind field is available at 12 m/s. The wind mill selected is of 10 m diameter and has rotate at 5 RPS with maximum efficiency of 35%. Find (a)available power density(b) the maximum power density (c) actual power density (d) power output from the wind mill (e) Axial thrust on the turbine **[07]**

****